

REVIEW ARTICLE

The Genoeconomics of Impulsive Intertemporal Choice: A Critical Review

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Abstract

Dr. Warren Bickel tirelessly investigated delay discounting (DD), a behavioral economic index of impulsive decision making, as a determinant of numerous health outcomes. Among the factors that contribute to a person's level of DD, a burgeoning body of research has illuminated its genetic foundations and the extent to which it may be a mechanistically informative phenotype for genetic contributions to addiction, obesity, attention-deficit/hyperactivity disorder, and a wide array of other health behaviors. In this review, we provide a concise history of quantification of DD as a human decision-making phenotype, tracing its origins from economics and operant learning theory, and critically synthesize the existing evidence on its genetic architecture. Finally, despite innate genetic contributions, we also discuss evidence that DD is a modifiable health behavior, with high potential in precision medicine.

INTRODUCTION

Delay discounting (DD), the extent to which an individual prefers smaller, immediate rewards over larger, delayed rewards, is a widely studied behavioral economic measure of impulsivity (Madden & Bickel, 2010), putatively reflecting a critical facet of self-regulation. It has been most extensively studied in the context of addiction and other mental health conditions (Bickel, Jarmolowicz, Mueller, Koffarnus, et al., 2012; Bickel & Marsch, 2001), reflecting a measurable preference for short-term gratification that undermines long-term goals, which frequently manifests in the decision-making patterns observed in compulsive drug use (De Wit, 2009), risky behavior (Donohew et al., 2000), and other adverse health behaviors (Stupiansky et al., 2013).

These transdiagnostic associations across a wide array of mental health outcomes have positioned DD as a

promising endophenotype—a heritable, quantifiable trait that mediates the relationship between risk- or protection-conferring genetic variation and health outcomes (MacKillop, 2013). Although delay discounting has traditionally been examined in relation to addiction and psychiatric disorders, its scope has expanded substantially in recent years. Steep discounting has been linked to a wide range of modifiable risky behaviors, including overeating (Amlung et al., 2016; Pearson et al., 2015), sedentary lifestyle (Phillips et al., 2020), poor sleep (Curtis et al., 2018; Law & Rasmussen, 2024; Massar & Chee, 2019), financial mismanagement (Hamilton & Potenza, 2012), sustainable climate action (Berry et al., 2017; Schleussner et al., 2024), and nonadherence to medical treatment (Barlow et al., 2016; Stupiansky et al., 2013). All of these, in turn, contribute to chronic diseases (Basu, 2025; GBD 2021 Risk Factors

In Memoriam of Warren K. Bickel – Contributions to Behavioral Economics, Decision-Making, and Addiction Science

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Collaborators, 2024) such as obesity, cardiovascular disease, and type 2 diabetes. This conceptual expansion has been further supported by evidence that DD shares neurobiological and genetic pathways with these outcomes (Parkes et al., 2019; Saeed et al., 2025; Thorpe et al., 2026; Vanes & Dolan, 2021), suggesting that it may act as a common upstream behavioral vulnerability. Thus, DD is increasingly studied not only as a measure of impulsivity but also as a biobehavioral bridge linking genetic liability to real-world health and social outcomes.

Over the past decade, advances in behavioral genetics have dramatically expanded the evidence base in this line of research. Progressing far beyond initial candidate gene studies, in particular, genome-wide association studies (GWAS) have revealed an emerging genetic architecture for DD and polygenic risk scores are illuminating how the genetic basis of DD is linked to liability for alcohol use, higher body mass index, tobacco use, and even the ability to successfully quit smoking. In the spirit of celebrating the life of Dr. Warren Bickel, a preeminent behavioral economist who tirelessly advanced the science of DD, this review traces its genetic foundations from initial small-scale and proof-of-concept observations to large-scale integrative genomic investigations. In addition to surveying the scientific landscape, we address future priorities in this area of research and consider DD's translational potential for personalized intervention.

THE EVOLUTION OF DELAY DISCOUNTING AS A PSYCHOLOGICAL PHENOTYPE

Scientific origins

Analysis of intertemporal choice in economics initially began with the development of the exponential discounting model in the first half of the twentieth century. The theoretical basis of this model assumes individuals display a time-consistent preference, that is, that discounting applies at a constant rate (Samuelson, 1937), quantified as

$$V = Ae^{-kD}, \quad (1)$$

where V represents the subjective value of the delayed reward, A is the objective amount of the delayed reward, e is Euler's number, D represents the delay, and the parameter k acts as a constant discounting rate. As k captures the steepness of the discount function, it is frequently the primary parameter of interest for quantifying individual differences in impulsivity.

Subsequently, however, discounting was shown to be time inconsistent rather than constant. This distortion in the valuation of delayed rewards led to it being described as impulsive (Ainslie, 1975) and later evolved into piceo-economic theory, which framed DD as a product of internal conflicts between present and future selves

(Ainslie, 1992). A critical insight from these models was that time-varying preferences implied preference reversals in which the subjective utility of an outcome qualitatively shifts based on time to receipt (Ainslie & Herrnstein, 1981). This in turn provides a quantitative model for the so-called "self-control failures" in which a person initially prefers a larger delayed reward but shifts their preference to a smaller immediate reward as it approaches in time. Furthermore, time inconsistency gave rise to the hypothesis that intertemporal choice is best understood using a hyperbolic (or hyperboloid) function, reflecting disproportionate devaluation of short-term delays and consistent devaluation of longer-term delays. One of the most common and well supported models for fitting discounting preferences (McKerchar et al., 2009) is Mazur's seminal one-parameter hyperbolic equation (Mazur, 1987):

$$V = \frac{A}{1 + kD} \quad (2)$$

Its widespread adoption was because the hyperbolic model, unlike the exponential model (Equation 1), includes a precipitous decline at short delays but permits the discounting rate to decrease as the delay gets longer, mapping more accurately to naturalistic choice patterns.

As the understanding of the mechanistic function of DD in humans evolved over time, so did the hyperbolic discounting equations where intertemporal discounting rate is represented by k (with higher values reflecting steeper discounting). Central to the discounting models introduced in Equation 1 and Equation 2 is the determination of indifference point whereby the immediate and delayed rewards are valued equally by an individual. Subsequent models to accommodate the human tendency to overevaluate shorter delays and underevaluate longer delays were developed using two-parameter equations of discounting by introducing a second free parameter to the hyperbolic equation (e.g., Myerson & Green, 1995; Rachlin, 2006). While model-based estimates to derive a k value provide a theoretical framework for discounting behavior, an atheoretical, data-driven approach is also widely used. The area under the curve (AUC) method (Equation 3; Myerson et al., 2001), leverages these indifference points by directly calculating the area beneath them and summing the area of the resulting trapezoids:

$$AUC = \sum_i (x_{i+1} - x_i) \left[\frac{y_i + y_{i+1}}{2} \right], \quad (3)$$

where x_i and x_{i+1} represent successive delays, y_i and y_{i+1} represent the corresponding indifference points at those delays, $i = 1, \dots, t$, and $t + 1$ is the number of indifference points. The AUC offers a robust measure that is less sensitive to nonsystematic data that might otherwise result in a poor model fit. Unlike k , where a higher value indicates more impulsivity, a lower AUC value represents

steeper discounting and thus higher levels of impulsive choice.

In animal and early human studies, operant tasks played a central role in examining delay discounting, typically involving repeated trials where subjects chose between a smaller sooner and a larger later reward (Mitchell, 2014). One influential example is the adjusting-delay procedure (Richards et al., 1999), which systematically varied the delay until subjects reached an indifference point. Conversely, the adjusting-amount procedure holds the delay constant while varying the reward size to reach indifference points (Moro, Saccenti, Seccia, et al., 2023; Richards et al., 1997). These titration methods allowed for a more granular estimation of individual discounting rates and enhanced the precision of real-time behavioral paradigms. Beyond titration methods, there are historically relevant procedures that do not rely on indifference points. For instance, the task developed by Evenden and Ryan (1996) presents subjects with choices between a smaller sooner and a larger later option, with the delay to the larger reward increasing across fixed blocks. In this paradigm, the primary dependent variable is the percentage of choices for the larger later option, providing a different perspective on how delay affects value in nonhuman subjects. These tools differ in format, reward type, and incentive structure, laying the groundwork for standardized instruments such as the monetary choice questionnaire (Kirby et al., 1999). The monetary choice questionnaire still leveraged the hyperbolic function of delay using Mazur's equation, but instead of fitting a nonlinear curve, the approximate location of indifference points was used, which extrapolates the k associated with this approximation.

Operationalizing delay discounting as a measurable trait

Along with novel quantitative modeling and assessments, foundational research has examined the validity of preferences expressed on delay discounting tasks. Numerous studies have validated the use of hypothetical rewards by comparing them with potentially real (Johnson & Bickel, 2002; Madden et al., 2004) or real rewards (Matusiewicz et al., 2013; Miller, 2019). In contemporary practice, some studies favor consequated tasks, using the offer of one of the reward options chosen at random, to ensure validity of preferences (Johnson & Bickel, 2002; Kirby et al., 1999; Madden et al., 2003; Mitchell, 1999), but most studies use hypothetical discounting tasks requiring participants to evaluate purely hypothetical rewards (e.g. (Kirby, 1997; Koffarnus & Bickel, 2014)). The test-retest reliability of delay discounting preference has also been extensively investigated (Anokhin, Golosheykin, et al., 2015; Smits et al., 2013), supporting its temporal stability (Kirby, 2009) and suggesting it can broadly be thought of as a trait (Odum, 2011a). While DD undergoes significant changes across development—

with steep discounting observed in childhood and adolescence, followed by a gradual decline that stabilizes in adulthood (Anokhin, Golosheykin, et al., 2015; Scheres et al., 2014; Steinberg et al., 2009)—this does not imply that DD is immutable over the life course (Odum, 2011b). Indeed, there is also evidence that robust experimental manipulations can increase or decrease DD (Bickel, MacKillop, et al., 2015; Scholten et al., 2019), but it is nonetheless considered a generally stable trait, *ceteris paribus*, and thus appropriate for investigating its genetic foundations.

Links to addiction and other psychiatric disorders

One of the most extensively studied domains of DD is its relationship with addictive behaviors. The theoretical basis is intuitive: Addictive disorders (e.g., substance use disorders; SUD) involve a preference for immediate rewards (e.g., intoxication) at the expense of longer term well-being, mirroring the discounting of delayed outcomes in DD paradigms (MacKillop, 2013). Systematic reviews and meta-analyses of substance use behaviors have found robust links between steeper DD and subclinical frequency and severity of substance use (Amlung et al., 2017) as well as the presence of an SUD (MacKillop et al., 2011). Although the magnitude of effect size varies, relations between steeper discounting and greater use/severity also appear substance agnostic, as reviews have found this association to be present for alcohol, tobacco, stimulants, opioids, and cannabis (Amlung et al., 2017; MacKillop et al., 2011; Strickland et al., 2021). Although limited in number, prospective studies have found higher DD in adolescence predicts future substance use and development of SUDs (Ferne et al., 2013; Fröhner et al., 2022; Peviani et al., 2024a, 2024b). However, not all studies find predictive validity. For example, Isen et al. (2014) found no prospective associations between DD and later substance use in a twin sample, and the association between steep DD and past substance use in males appeared to be mediated by cognitive ability. Such findings reinforce the view that impulsivity and executive control are intrinsically linked as opposing constructs (Bickel, Jarmolowicz, Mueller, Gatchalian, et al., 2012; Bickel, Moody, et al., 2014). This dual-process framing positions DD as a tractable behavioral measure of self-regulation, relevant not only for understanding addiction but also for a broader range of mental and physical health outcomes.

An expanding body of literature implicates DD as a transdiagnostic marker for psychiatric disorders. In a meta-analysis examining DD and several psychiatric disorders, Amlung et al. (2019) found that steeper DD was significantly associated with major depressive disorder (MDD), bipolar disorder, borderline personality disorder, bulimia nervosa, and binge eating disorder. Other reviews have extended this evidence to include posttraumatic stress

disorder (Bird et al., 2024) and attention-deficit/hyperactivity disorder (ADHD; Jackson & MacKillop, 2016). Although deficits in self-control are core features of ADHD, like addiction, the underlying mechanisms that explain relationships with DD may differ for other psychiatric conditions. For example, DD can be conceptually connected to an excessive present focus in posttraumatic stress disorder (Bird et al., 2024) and steeper DD in individuals with MDD has been hypothesized to be related to hopelessness about the future, and therefore preference for more immediate rewards (Pulcu et al., 2014). In Amlung et al. (2019), shallower DD was consistently negatively associated with anorexia nervosa, which can be conceptualized as a reflection of the maladaptive high levels of self-control present in the disorder (E. Levitt et al., 2020). In the personality disorder domain, while links with borderline personality disorder and antisocial personality disorder represent the “undercontrolled” expressions akin to addiction and ADHD (Moody et al., 2016; Paret et al., 2017), individuals with obsessive-compulsive personality disorder exhibit abnormally low discounting rates, akin to overcontrol observed in anorexia nervosa (Pinto et al., 2014). These suggest that self-regulatory capacity functions as a behavioral spectrum, where clinical dysfunction arises at both extremes (Figure 1). This then allows us to conceptualize DD as a robust transdiagnostic factor that bridges traditional diagnostic categories. Indeed, two studies that have evaluated discounting specifically as a

transdiagnostic indicator affirm its utility across many psychiatric conditions but importantly highlight that there are nuances and may be important latent intermediary mechanisms that warrant future research (Keidel et al., 2024; E. E. Levitt et al., 2023).

Collectively, based on these diverse links, DD has been identified as promising indicator (Lempert et al., 2019) within the research domain criteria (RDoC) initiative (Insel et al., 2010) of the National Institute of Mental Health, which seeks to identify dimensional psychological and neurobiological processes that transcend traditional diagnostic boundaries. Within the RDoC framework, DD is considered a promising paradigm for capturing deficits in cognitive control, reward processing, and temporal decision making that manifest across multiple disorders.

Links to other health behaviors

DD has also been investigated in relation to general health behaviors and outcomes. One systematic review found that steeper DD, whether measured through money or through food, was robustly associated with obesity (Amlung et al., 2016). This is consistent with the conceptualization of obesity as a disorder mechanistically similar to addiction, with more impulsive behavior leading to overconsumption (Bickel et al., 2021; Volkow & Wise, 2005). Other domains of health have also been

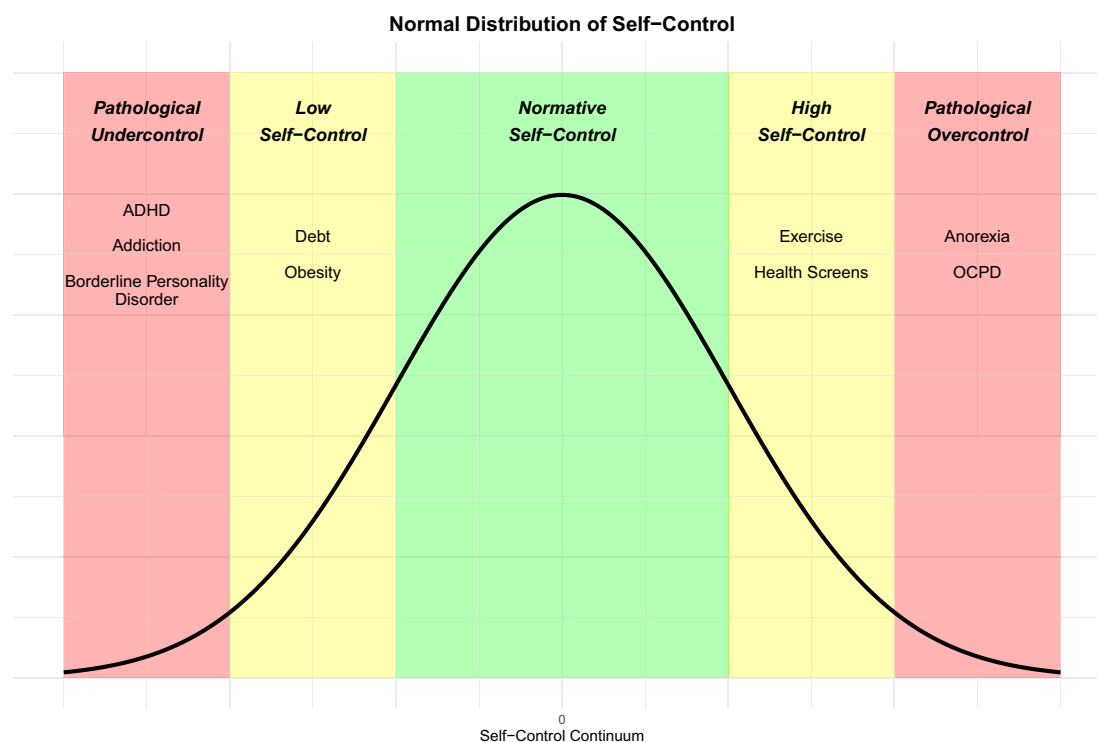


FIGURE 1 Self-regulatory capacity as a behavioral spectrum with clinical dysfunction emergent at both extremes. This figure illustrates the continuum of self-control as a normal distribution. Colored regions indicate different levels of self-regulation capacity: The green zone represents normative self-control, yellow zones indicate low and high self-control with associated behavioral patterns, and red zones represent pathological extremes of undercontrol and overcontrol. Examples of conditions and behaviors associated with each zone are listed vertically in alphabetical order.

examined DD has been associated with compulsive eating (i.e., “food addiction,” Minhas et al., 2021), sexual risk behavior (Jones et al., 2018; MacKillop, Celio, et al., 2015; Reimers et al., 2009), physical activity and sedentary behavior (Cabral et al., 2024; Daugherty & Brase, 2010; Wang et al., 2024), and primary and secondary prevention health behaviors, such as wearing a seat-belt, wearing sunscreen (Daugherty & Brase, 2010), and cancer screening (Bradford, 2010; Sheffer et al., 2018). Emerging research also links social media engagement with steeper DD in both young and older adults (Augner et al., 2023; Ding et al., 2018; Schulz van Endert & Mohr, 2022; Su et al., 2024). These numerous and diverse links can be framed as the manifold ways that trade-offs between immediate gratification (or effort avoidance) and longer term health benefits are present across health behaviors.

GENETIC BASIS OF DELAY DISCOUNTING

The investigation of DD as a genetically influenced trait has evolved alongside major advances in genetic studies. Early insights came from animal models, where strain-based differences in impulsive choice suggested a heritable component. These findings paralleled emerging family studies and later twin studies. In the early 2000s, research shifted toward candidate gene

studies (Kwon & Goate, 2000), primarily focusing on dopaminergic and serotonergic pathways. Around this time, genome-wide genotyping technology was becoming available, but cost and data availability limited its use. Only with the expansion of large-scale population cohorts and biobank infrastructure in the 2010s did genome-wide association studies (Hirschhorn & Daly, 2005) become feasible for DD. Existing standardized computer programs and brief assessment protocols to administer DD, with the outcome typically operationalized as the log-transformed discounting rate ($\log k$), were also central to this transition to large-scale genomic studies. See Figure 2 for a timeline of the major developments in human quantitative and molecular genetic studies.

Oblique evidence from animal and family studies

Inbred rodent strain studies, which allow genetic differences to be isolated from environmental factors (Beck et al., 2000), have consistently shown systematic discounting differences across strains. Early work found higher discounting in Lewis versus Fischer 344 rats (Anderson & Woolverton, 2005), which was replicated in several subsequent studies (Madden et al., 2008; Stein et al., 2012). A sizeable variation in DD task performance has been attributable to genetic factors in both rat (Richards et al., 2013) and mice models (Isles et al., 2004).

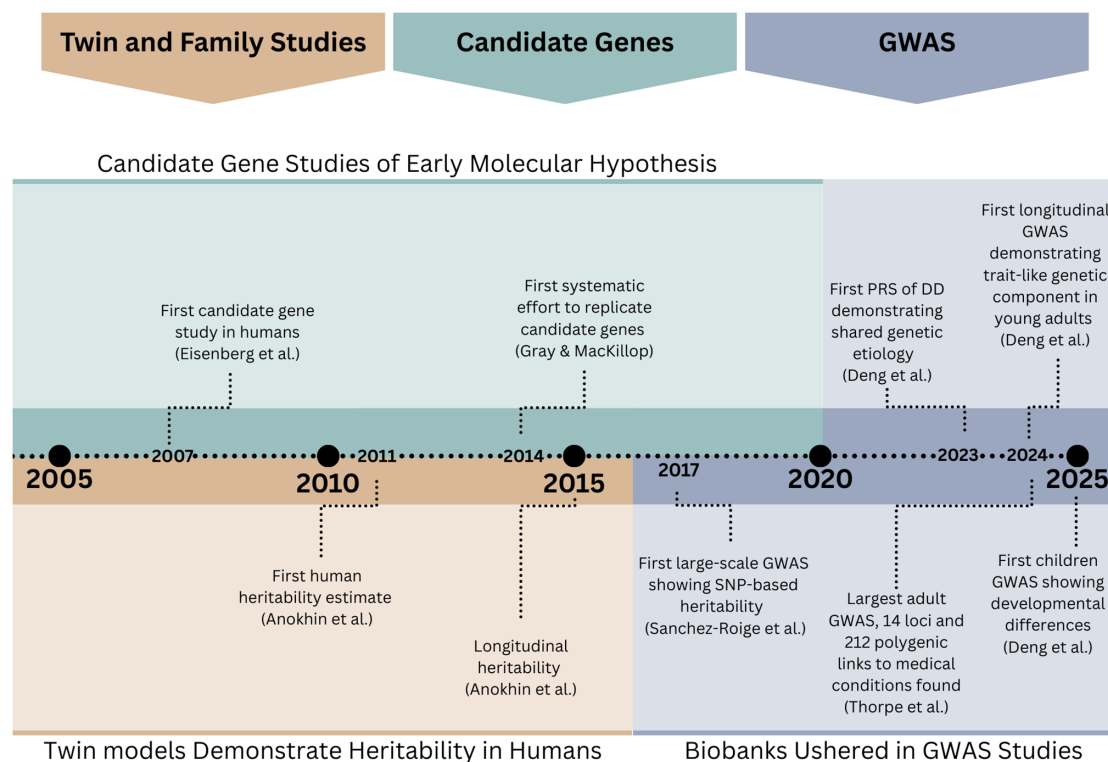


FIGURE 2 Timeline of key developments in human quantitative and molecular genetic studies of delay discounting. Events are color-coded by study type: twin and family studies (orange), candidate gene studies (green), and genome-wide association studies (GWAS; blue).

Similarly, much of the early evidence for familial transmission of impulsive choice comes indirectly through research linking family history of SUDs with elevated DD. These studies do not typically assess intergenerational concordance in DD but instead hypothesize that steeper discounting may represent an intermediate phenotype through which inherited risk for addiction is expressed. This line of research has shown that individuals with a positive family history of addiction tend to discount future rewards more steeply than those without such a history (Acheson et al., 2011; Athamneh et al., 2017; Dougherty et al., 2014; VanderBroek et al., 2016). Although this design cannot disentangle environmental consequences of family history of addiction (e.g., parenting effects, psychosocial adversity), they nonetheless imply DD may be a mechanism for genetic influences.

Quantitative evidence from twin studies

The classical twin design is often used to partition the contribution of genes and environment to trait variability (Boomsma et al., 2002) by contrasting the trait resemblance within monozygotic and dizygotic twin pairs. The most cited estimates of DD heritability come from adolescent twin populations, with values reported between 30% and 50% (Anokhin et al., 2011; Anokhin, Grant, et al., 2015; Isen et al., 2014). Importantly, DD heritability appears to vary across development. In midchildhood, heritability of DD was estimated at 25% (Morais et al., 2025). In early adolescence (ages 12 to 14), heritability rises from ~30% to ~50% (Anokhin et al., 2011) and continues to increase, with estimates of 35% and 55% at age 16 and 18 (Anokhin, Grant, et al., 2015), respectively. These changes likely reflect developmental changes in prefrontal systems associated with self-control and reward valuation (Friedman & Robbins, 2022). Notably, these studies rely on single-item scenarios or adjusted-amount-titration procedures, which, despite their brevity, captured meaningful individual differences and offered some of the earliest robust evidence supporting DD as a candidate endophenotype. Despite this, these designs cannot identify specific genes responsible for DD, reinforcing the need for molecular follow-up.

Initial molecular genetic studies: The (refuted) dopamine hypothesis

Candidate gene studies emerged in the early 2000s as an initial attempt to address the limitations of twin and family-based studies. Several neurotransmitter systems have been examined in relation to DD, with most research focused on dopaminergic pathways. This was based on extensive preclinical research implicating dopamine in reward valuation (Hollerman & Schultz, 1998;

Kobayashi & Schultz, 2008). Among these, the most frequently studied genes include the NTAD (*NCAMI-TTC12-ANKK1-DRD2*) gene cluster (Mota et al., 2012), which has been linked to altered dopamine receptor availability (Pohjalainen et al., 1998; J. Thompson et al., 1997); *COMT*, which modulates prefrontal dopamine tone (Dreher et al., 2009; Meyer-Lindenberg et al., 2005); and *DRD4*, which has been shown in mouse models to regulate striatal glutamate signaling (Thomas et al., 2009). *DRD1* and *DRD3* have also been implicated, but it has been less frequently examined. Table 1 summarizes all published candidate gene studies for DD in human samples.

The first studies focused on polymorphisms related to the dopamine D2 receptor gene. The 'TaqIA' polymorphism (rs1800497), originally thought to be part of *DRD2* but now known to reside in the adjacent *ANKK1* gene, has been associated with steeper DD in individuals with European ancestry (Eisenberg et al., 2007; MacKillop, Gray, et al., 2015) but not in Japanese (Kawamura et al., 2013). Another SNP in *DRD2* (rs6277) was shown to associate with faster response but not steeper discounting (White et al., 2009). Two other dopamine-related genes, *COMT* and *SLC6A3* (*DAT1*), have been moderately linked to DD (Boettiger et al., 2007; MacKillop, Gray, et al., 2015; Paloyelis et al., 2010) showing the Val carriers to exhibit steeper DD. In particular, Paloyelis et al. (2010) found that both the *COMT* Val158Met polymorphism and the *DAT1* 10/6 haplotype predicted steeper DD in a hypothetical reward task but not in a real-time task in adolescents with and without ADHD. In addition, Smith and Boettiger (2012) showed that the influence of *COMT* on DD varies with age such that discounting decreased with age in Met carriers but increased in Val/Val individuals. However, the *COMT* variant association was not replicated in users of heroin (Moses et al., 2020).

The dopamine D4 receptor gene, particularly the 7-repeat allele, has received attention in the impulsivity literature, but findings regarding DD have been inconsistent. While the 7R allele has been associated with steeper DD in carriers with low socioeconomic status versus those with high socioeconomic status (Sweitzer et al., 2013), two other reports found no significant association (Eisenberg et al., 2007; Paloyelis et al., 2010). Of interest, Eisenberg et al. (2007) found an epistatic interaction such that carriers of both the risk allele in rs1800297 linked to *DRD2* and 7-repeat allele exhibited dramatically more impulsive discounting, although this finding has not been replicated. Dopamine receptor genes *DRD1* and *DRD3* have received the least attention in DD research to date, but emerging findings suggest their potential relevance in addiction-related impulsivity. In a study of 106 individuals who actively use heroin, Moses et al. (2020) found that the *DRD1* rs686 variant modulated DD behavior across satiation and withdrawal states, pointing to a role for *DRD1* in state-dependent

TABLE 1 Human molecular candidate genetic studies of delay discounting.

Study	Sample*	Gene / Polymorphism	Key Findings
Eisenberg, MacKillop et al. (2007)	195 university students	ANKK1/DRD2 Taq1A (rs1800497) and DRD4 48 bp VNTR	TaqI A1+ carriers showed steeper DD; significant interaction between A1 carriers and DRD4 long allele carriers
Boettiger et al., (2007)	19 healthy adults	COMT(val158met; rs4680)	Val carriers exhibit steeper DD.
White et al., (2009)	72 young adults	DRD2 C957T	C allele was associated with faster response but not steeper DD.
Paloyelis et al., (2010)	68 boys (ADHD-CT = 36; control = 32) aged 11–20 years	COMT(val158met; rs4680) and DRD4(48 bp-VNTR) polymorphisms	DAT1 and COMT variants associated with impulsivity and steeper discounting in a hypothetical but not real-time task; magnitude effects specific to ADHD-CT group
Smith & Boettiger (2012)	142 late adolescent and adults	COMT(val158met; rs4680)	Among met-carriers, DD was negatively correlated with age from late adolescence to adulthood but positively correlated among val-carriers.
Sweitzer et al., (2013)	546 community adults	DRD4 exon 3 VNTR	Steeper DD in 7-repeat carriers from low-SES backgrounds vs. high-SES
Kawamura et al., (2013)	91 healthy Japanese adults	C957T (rs6277) and TaqI A (rs1800497) in DRD2	Individuals with minor alleles (T) showed more impulsive discounting. No association for TaqI.
Gray and MacKillop (2014)	175 weekly gamblers	Candidate dopamine-related genes	Results do not replicate previous associations but identify several novel candidates.
MacKillop, Gray, et al. (2015)	713 daily smokers	rs4680 in COMT, rs1800497 in ANKK1, and three haplotypes proximal to rs1800497	Risk allele carriers exhibited higher discounting at the SNP-level
Moses et al., (2020)	106 current users of heroin	DRD1 rs686 and rs5326, DRD3 rs6280, COMT rs4680	DRD1 rs686 variant significantly modulates heroin DD differences between satiation and withdrawal states and thus associated with increased drug-use impulsivity

*White European unless other than specified.

impulsive decision making. Notably, a systematic replication effort by Gray and MacKillop (2014) examining multiple dopamine-related genes in a sample of 175 gamblers did not replicate previous associations with rs1800497 and rs4680, although nominal associations were detected for variants in *DRD3*, *SLC6A3*, *DDC*, *DBH*, and *SLC18A2*. These inconsistent findings may reflect methodological limitations, including small sample sizes, population stratification, and reliance on single-marker analyses.

This reinforced concerns about false positives and emphasized the need for larger, better-powered designs. On balance, although early molecular genetic studies ultimately yielded inconsistent findings, there remains potential value in using a systems biology approach, particularly within large-scale carefully curated samples, to examine specific DD processes. Further, these findings align with recent PET imaging and meta-analytic evidence showing minimal associations between D2 receptor availability and DD in healthy individuals but stronger, often inverse associations in clinical populations with altered dopamine function (Ballard et al., 2015;

Castrellon et al., 2019). These results suggest that dopamine-related genetic effects on DD may differ systematically between healthy and clinical groups, underscoring the importance of studying DD across diverse neurobiological states rather than in healthy volunteers alone.

Genome-wide association studies: A series of firsts

Over the past decade, the field of behavioral genetics has shifted its focus toward GWASs and polygenic frameworks, overcoming many of the limitations of earlier candidate gene approaches to better capture the genetic architecture. Large-scale GWAS have transformed our understanding of DD (Table 2) by identifying robust loci, enabling the construction of polygenic risk scores (PGS) for cross-phenotype prediction, and revealing genetic correlations that suggest shared liability with substance use, externalizing behavior, psychiatric and cognitive traits, and late-onset chronic diseases.

TABLE 2 Genome-wide association studies of delay discounting in humans.

Study	Sample*	Key findings
Sanchez-Roige et al., (2018)	23,217 adults from 23andMe	The first large-scale GWAS to demonstrate SNP-based heritability of 12% and genetic overlap with other traits.
MacKillop et al. (2019)	986 healthy young adults	Results not supportive of the previous candidate gene studies and provide initial evidence of complex genetic architecture.
Deng et al. (2023)	1,534 community adults	First polygenic study of DD that demonstrate shared genetic etiology and reinforcing DD's place in the externalizing and impulsivity spectrum.
Thorpe et al. (2026)	134,935 adults from 23andMe (an update from Sanchez-Roige et al., 2017)	Largest adult GWAS to date, identifying 14 significant loci, widespread genetic correlations with behavioral, physical, and neuroimaging traits, and polygenic links to 212 medical conditions, supporting DD as a pleiotropic and polygenic endophenotype for health.
Deng et al., (2025)	3,860 healthy young adults	First longitudinal GWAS that demonstrates a stable trait-like genetic component in young adults and increases in signal via multiple observations over time.
Deng et al., (2025)	2,897 children from ABCD	First children GWAS that shows both continuity and developmental differences in DD genetic architecture.

The first well-powered GWAS of DD was conducted by Sanchez-Roige et al. (2018), leveraging population-level data from over 23,000 participants of European ancestry collected through 23andMe, a consumer-oriented, self-initiated genetic testing platform. This is the first robust evidence that DD has a polygenic basis, meaning the trait is influenced by many genetic variants, each contributing only a small effect. Together, they account for roughly 12% of the variation in DD. This estimate, also known as the chip-heritability, reflects the proportion variance explained by common genetic variants measured on genomewide arrays (Visscher et al., 2008). This study also identified a genomewide significant association (rs6528024; intron of *GPM6B*) on the X chromosome but has not been replicated. Another GWAS was conducted in a smaller sample of 986 healthy and substance-free young adults (MacKillop et al., 2019), finding support for one a priori locus (rs521674 in *ADRA2A*) and detecting one genomewide significant association (rs13395777), but in an intergenic region of unknown function. While these initial GWAS results were modest in effect size, they marked a shift away from underpowered candidate gene studies and toward data-driven discovery.

Recent innovations include the first longitudinal GWAS of DD, Deng et al. (2025) leveraged repeated measures in 3,860 young adults across three ages (aged 24, 26, and 28), finding that DD demonstrates moderate-to-high temporal stability and a chip-based heritability of 22%. These results reinforce the conceptualization of DD as a stable, trait-like construct and highlight the potential utility of genetically informed longitudinal designs for understanding individual trajectories of impulsive decision making. The largest effort to date, Thorpe et al. (2026), analyzed DD discounting in an expanded 23andMe cohort of 134,935 participants, with improved statistical power to identify genetic associations. This study identified 14 genomewide significant loci, but most

lead SNPs—except for one—were located in regions previously associated with a wide range of behavioral and psychological traits in prior GWASs. For example, the top hit (rs34645063) lies between *MMS22L* and *POU3F2* and is linked to traits such as risk-taking tendencies, substance use, psychiatric disorders, cognitive ability, and body mass index. However, none of the genomewide significant genes identified have a well-established biological function directly linked to impulsive choice, and the associated loci span diverse pathways without converging on a specific biological system.

In parallel with adult studies, the first GWAS for children was conducted in a subset of 2,897 participants from the Adolescent Brain Cognitive Development Study (Deng, Elsayed, et al., 2025), providing a developmental lens. This subset was defined based on phenotypic quality control screening and European ancestry status. Chip-based heritability estimates were modest (~20%) but slightly larger than those reported in large samples of adults (10–12%). Genetic correlations between child and adult were modest ($r = .26$), yet genetic scores derived from adult GWAS were significantly associated with DD in children, underscoring both continuity and developmental differences in its genetic architecture. These increasingly sophisticated GWAS studies represent a turning point for understanding the genetics of DD, providing stronger support for trait-relevant discoveries and opening the door to integrative post-GWAS analyses.

Post-GWAS analyses and insights from bioinformatics

Post-GWAS approaches (Gallagher & Chen-Plotkin, 2018) extend initial GWAS findings through additional analyses, such as functional analysis, genetic correlation studies, PGS, and genomic structural equation modeling (gSEM; Grotzinger et al., 2019). These analyses have

advanced our understanding of DD's genetic architecture and highlighted its transdiagnostic properties through shared genetic foundations with other traits.

Genetic correlation analyses (Deng, Belisario, Munafò, et al., 2025; Sanchez-Roige et al., 2018; Thorpe et al., 2026) show common genetic influences with a range of behavioral, cognitive, and clinical traits. DD is positively genetically correlated with SUDs, ADHD, obesity, type 2 diabetes, coronary artery disease, risk-taking behaviors (e.g., high-risk sexual behavior), and several brain measures (Sanchez-Roige et al., 2018; Thorpe et al., 2026), supporting its role as a transdiagnostic indicator of impulsivity and self-regulation deficits. Conversely, DD shows negative genetic correlations with educational attainment and cognitive ability, suggesting that greater impulsive choice is linked to lower executive functioning and long-term planning capacity. These findings position DD within a broader externalizing spectrum and are consistent with the reinforcer pathology model (Bickel et al., 2011; DeHart et al., 2020; Jarmolowicz et al., 2016), which frames DD as a core mechanism underlying maladaptive reward processing and also at the intersection of cognitive control and impulsive tendencies. Local genetic correlation analyses further highlight specific loci, such as *chr11q23.1* (spanning the *NCAM1-TTC12-ANKK1-DRD2* cluster), that are responsible for the shared genetics between DD and addiction risk (Thorpe et al., 2026), aligning with some of the findings from early candidate gene studies.

PGS analyses have also been central to post-GWAS efforts in understanding DD across three main paradigms. Early PGS research on DD began by leveraging aggregated genetic scores from other traits to predict DD ($\text{PGS}_{\text{Other Trait}} \rightarrow \text{DD}$) at a time when well-powered GWAS of DD were not yet available. These studies found that polygenic risk for conditions such as ADHD, smoking, body mass index, and education attainment was associated with DD, suggesting shared genetic etiology and reinforcing DD's place in the externalizing and impulsivity spectrum (Deng et al., 2023). With the emergence of larger DD GWASs, two additional paradigms became feasible. First, studies using PGS for DD to predict DD itself ($\text{PGS}_{\text{DD}} \rightarrow \text{DD}$) confirmed that aggregated genetic risk significantly explains individual variability in impulsive choice, validating DD as a genetically influenced trait (Deng, Belisario, Munafò, et al., 2025; Deng, Elsayed, et al., 2025). Second, cross-trait predictions using PGS for DD to predict other outcomes ($\text{PGS}_{\text{DD}} \rightarrow \text{Other Traits}$) demonstrated associations with substance use, externalizing behaviors, ADHD symptoms, gambling, and lower educational attainment (Deng, Belisario, Doggett, et al., 2025; Rabinowitz et al., 2025; Thorpe et al., 2026), further supporting DD's relevance as a transdiagnostic vulnerability factor. These three paradigms established DD's role as both an outcome and a driver of broader neurobehavioral risk, reinforcing its utility in dimensional models of addiction.

Genomic structural equation modeling uses GWAS summary statistics to estimate the shared and unique genetic architecture across multiple traits. This approach has gained success in identifying latent genetic factors underlying dimensions that explain how different traits are genetically related. For example, it has been shown that DD was more strongly genetically correlated with a latent genetic factor for SUDs than general substance use and remained independently correlated with the SUD genetic factor even after accounting for shared genetic variance (Vilar-Ribó et al., 2025). This highlights DD as being specifically linked to the risk for SUDs rather than general experimentation or use, supporting its role in addiction liability models (MacKillop, 2013). Further, a recent gSEM study identified three genetically distinct facets of impulsivity, lack of self-control, reward drive, and sensation seeking, each linked to differential brain development patterns in children aged 9–14 (Shishikura et al., 2025). Complementary work used confirmatory gSEM to study the dual-systems model, demonstrating top-down and bottom-up impulsivity traits capture distinct but related sources of genetic variance, with differential associations to brain morphometry in adults (A. P. Miller & Gizer, 2024). These highlight the neurodevelopmental expression of impulsivity-related genetic risk and further support the neurobiological basis of DD.

While most genomic studies of DD have focused on additive effects, emerging work highlights the importance of interaction models, including gene $\times$ gene (epistasis) and gene $\times$ environment effects. For example, early candidate studies suggest that dopaminergic gene combinations (e.g., *COMT* $\times$ *DAT1*, *ANKK1* $\times$ *DRD4*) and environmental moderators like socioeconomic status or early childhood adversity may shape how genetic risk for impulsive choice manifests across development and contexts. In more recent studies, the polygenic impulsivity dimension for lack of self-control also showed moderation by socioeconomic status to influence white matter mean diffusivity (Shishikura et al., 2025). Similarly, in a recent analysis of the Avon Longitudinal Study of Parents and Children cohort, DD polygenic scores were modestly associated with smoking behavior among young adults, with effects moderated by sex (Deng, Belisario, Doggett, et al., 2025), highlighting how sex differences may shape the expression of DD-related genetic liability. Nevertheless, these remain underexplored in GWAS-era DD research and represent an important frontier for understanding individual differences in DD.

ACTIVE RESEARCH PRIORITIES

DD now meets many criteria for a transdiagnostic endophenotype, supported by converging genetic, neurobiological, and behavioral evidence. The field has entered a new phase focused on integrating molecular and neural

markers, testing causality, expanding diversity, and developing targeted interventions. Active research continues to refine the conceptualization of DD as both a trait and intervention target and to clarify its temporal role in the onset and maintenance of psychiatric disorders and health behaviors more broadly.

Genetic interpretability and inclusivity

Despite progress in identifying genetic correlates of DD, fundamental limitations constrain both theoretical understanding and practical applications. Most statistically significant genes identified in DD genetics lack well-characterized biological functions, while genes with well-established neurobiological relevance often fall short of genomewide significance. This reflects a critical gap in the field: Statistical associations remain largely disconnected from biological mechanisms. The biological roles of associated loci are poorly understood, particularly in relation to neurotransmitter signaling, neural circuitry, and brain development. Parallel advances in brain imaging genomics (Elliott et al., 2018; Mascarell Maričić et al., 2020; S. M. Smith et al., 2021; P. M. Thompson et al., 2014; B. Zhao et al., 2019) highlight neurobiological substrates that may underlie genetically influenced impulsive choice but remain underused. Similarly, efforts to refine the DD phenotype might improve genotype–phenotype associations, for example, through a better delineation of component processes (Mitchell, 2019). However, this requires assessment tools that can account for processes related to reward size, delayed period, and available alternatives and additional processes not hypothesized (Marshall et al., 2014). Thus, our current understanding of the DD genetic architecture remains at best fragmented and opaque, limiting the ability to map statistical signals onto plausible neurobiological pathways or therapeutic targets.

Critically, genetic research of DD needs to expand beyond Western, educated, industrialized, rich, and democratic populations, with increasing attention to cross-ancestry genetic generalizability, sociocultural moderators, and global burden of impulsivity-linked health outcomes. For other traits, previous studies have demonstrated that genetic associations can be highly sensitive to ancestral background and environmental context (Abdellaoui et al., 2023; Mahajan et al., 2022; Ntzani et al., 2012). This limitation, common in GWAS and post-GWAS studies of other complex human traits, contributes to health disparities across populations with different backgrounds (Martin et al., 2019). Cultural, socioeconomic, and gender-based variability in DD patterns remain underexplored and thus raise questions about the cross-ancestral generalizability of current findings and potential gene $\times$ environment interaction effects.

Leveraging DD genetic instruments in novel data sets and to inform causality

Although discussed earlier, it is worth briefly noting the growing utility of DD PGS as proxy measures in large-scale cohort and biobank studies where direct DD phenotyping is unavailable or no longer feasible. Genetic instruments can enable retrospective and scalable analyses, allowing researchers to investigate the downstream effects of DD in already collected data. For example, DD PGS have been linked to a broad spectrum of addiction-related outcomes in phenome-wide association studies (Deng, Belisario, Doggett, et al., 2025; Thorpe et al., 2026). Thus, DD can be explored in broader domains of preventive medicine and public health using a genetic instrument.

Further, although epidemiological studies suggest a bidirectional relationship between DD and addiction, the causal nature of these associations remains unresolved. Causal inference methods, most notably Mendelian randomization (MR), offer a powerful framework to address this question (Richmond & Smith, 2022). Specifically, MR leverages genetic variants as instrumental variables, with risk alleles randomly allocated at conception in a manner analogous to treatment assignment in randomized controlled trials. This natural randomization allows MR to test whether an observed association between an exposure (e.g., DD) and an outcome (e.g., SUD) is consistent with a causal effect rather than confounding or reverse causation. Applied to DD, MR could clarify whether genetically influenced tendencies toward steep DD contribute to the development of addiction, other psychiatric disorders, or other health outcomes, such as obesity or cardiovascular disease. To date, however, such approaches have not been widely applied in DD research, likely because of insufficiently strong effects to serve as robust single instruments. Instead, polygenic instruments, aggregating information across many variants of small effect, may provide sufficient power to conduct MR analyses (Dudbridge, 2021), making them a promising next step for causal inference in this domain.

Delay discounting as a modifiable clinical target

Although DD is heritable and exhibits trait-like stability over time (Odum, 2011a, 2011b), it is also proven to be at least partially modifiable (Koffarnus et al., 2013; Rung et al., 2019). This duality makes DD particularly compelling as a clinical target. Reconciling genetic influences with intervention strategies can be illustrated by the well-established example of cardiovascular disease and a genetically causal risk factor, elevated low density lipoproteins (LDL), as demonstrated by multiple lines of causal evidence from longitudinal and MR studies (Ference et al., 2017). For example, an individual may have a genetic propensity for higher LDL-C across the

lifespan. This represents a stable, higher risk phenotype, yet LDL-C levels remain modifiable through interventions such as diet or statins (Yusuf et al., 2004), which can be tailored to address the high genetic risk (Livingstone et al., 2021). By analogy, individuals may be predisposed to steeper DD, showing greater immediate reward orientation, but DD is also somewhat malleable, at least in response to targeted interventions. In this sense, genetic propensity for higher DD or other complex, multifactorial traits, is not deterministic but reflects a probabilistic liability that can be targeted and modified. Extending this logic, knowledge of an individual's genetic risk (e.g., DD-PRS) or behavioral phenotype (e.g., norm-referenced testing) could be leveraged in personalized medicine frameworks, whereby individuals with higher DD selectively receive interventions designed to extend their future time horizons.

DD as a modifiable clinical target is well supported, but the efficacy of current strategies remains distinct across modalities. Initial behavioral approaches often focused on impulse control training, which was designed to strengthen inhibitory mechanisms via the concepts of delay fading (Mazur & Logue, 1978), delay of gratification (Burns et al., 2021; Mischel et al., 1989), or systematic exposure to delay (Renda et al., 2018). However, this body of literature has proven variable (Friese et al., 2017), with many studies reporting inconsistent findings regarding the long-term maintenance and generalizability of training effects (Rung & Madden, 2018). In contrast to the limitations of inhibitory training, emerging learning-based techniques and cognitive strategies such as episodic future thinking show significant promise in producing more robust reductions in DD regardless of the emotional valence of the future event (Ballance et al., 2022; Bulley et al., 2020; Snider et al., 2016). Consistent with the preceding personalized medicine hypothesis, episodic future thinking has been found to have a rate-dependent effect such that it disproportionately ameliorates DD for the most impulsive individuals (Shevorykin et al., 2021; Snider et al., 2018). These findings more broadly converge with the hypothesis that malleability varies based on the extremity of the phenotype (Bickel, Landes, et al., 2014; Bickel, Quisenberry, et al., 2015). Thus, interventions to address DD may be viable but are likely to be specifically useful for individuals exhibiting objectively high immediate reward valuation, based on either innate genetic risk or acquired environmental causes. Literature has also identified more intensive approaches to modify DD including pharmacological approaches (Perkins & Freeman, 2018) as well as noninvasive neuromodulation targeting regions such as the orbitofrontal cortex (Moro, Saccenti, Ferro, et al., 2023) and the dorsolateral prefrontal cortex (Amlung et al., 2014; He et al., 2025; Kekic et al., 2017; Owens et al., 2019). These regions represent established therapeutic targets across multiple psychiatric conditions, including MDD (Y.-J. Zhao et al., 2025), schizophrenia (Wen et al., 2021), and

SUDs (Mehta et al., 2024), which emphasize the clinical potential of targeting these hubs to reduce impulsive decision making in populations with psychiatric disorders. Furthermore, as noninvasive brain stimulation has been shown to improve long-lasting synaptic plasticity (Huang et al., 2005), a combined strategy involving both behavioral interventions and neuromodulation has been recommended as a potentially more powerful approach to rehabilitate DD (Saccenti et al., 2024). Collectively, work in this area reframes DD as not only an etiological process in health behavior but also a promising treatment target among those exhibiting pathological levels.

CONCLUSION

The accumulating literature on genetic influences on DD preferences enrich a multifaceted conceptual and experimental understanding of this fundamental human decision-making phenotype. Rather than being a departure from the original economic assumptions or operant principles, this work illuminates mechanistic substrates that help explain stable between-subject variability in responsiveness to delayed consequences. In other words, understanding DD's genetic foundations offers a deeper account of individual differences in intertemporal reward preferences and the associated health-related outcomes. The integration of genetic methods has also opened new avenues, including the identification of shared biological pathways linking DD with addiction, obesity, and chronic diseases and the development of precision interventions informed by an individual's genetic liability to impulsive choice. Translational efforts now seek not just to measure DD, but to change it in enduring ways toward meaningful clinical benefits. From this wider perspective, DD serves as a key unifying construct across several disciplines, from genetics, behavioral pharmacology, and developmental psychopathology to behavioral economics, intervention research, and public health. The continued convergence of genetic, neural, and behavioral approaches affirms DD as a central focus for interdisciplinary inquiry in the science of both normative and maladaptive behavior.

AUTHOR CONTRIBUTIONS

WQD and JM conceived and designed the review. JM acquired funding. WQD, KLB, AD, ME, AC, and JM contributed to drafting the manuscript. WQD and JM drafted Sections 1 and 5, KLB drafted Sections 2.1 and 2.2, AD and ME drafted Sections 2.3 and 2.4, WQD drafted Section 3, and AC and WQD drafted Section 4. WQD prepared the tables. JM prepared Figure 1, and KLB prepared Figure 2. All authors critically reviewed and revised the manuscript and approved the final version.

CONFLICT OF INTEREST STATEMENT

JM is a principal in Beam Diagnostics, Inc. No other authors have disclosures.

DATA AVAILABILITY STATEMENT

No new data were generated or analyzed in this review. All information discussed was obtained from publicly available published sources, which are cited in the reference list.

ETHICS APPROVAL

This review article synthesizes information from previously published studies and does not involve the collection of new data from human participants, animals, or identifiable personal information. Research ethics board approval was not required.

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